

Embedded Systems with STM32

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Preface

This book is an introduction to embedded systems programming using an STMicroelectronics STM32 microcontroller.

i Draft status

This book is a work in progress. Chapter content, the specific STM32 part number/development board, and toolchain (e.g. STM32CubeIDE) used throughout the book still need to be filled in.

It is intended for readers who are new to embedded systems, and covers the fundamentals of microcontrollers, C programming for resource-constrained targets, general purpose input/output (GPIO), embedded programming patterns, timers, analog-to-digital conversion, and serial communication.

How this book is organized

The book is split into parts, each grouping related chapters:

- **Foundations** — what an embedded system is, and the C programming concepts needed to work with one.
- **Digital I/O and Program Structure** — controlling GPIO pins and structuring larger embedded programs (hardware abstraction layers, state machines, interrupts).

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- **Peripherals** — timers, analog-to-digital conversion (ADC), and serial communication.

Relationship to the MSP430 edition

This book follows the same chapter structure as the companion “Embedded Systems with the MSP430G2553” textbook, but is written independently for the STM32 platform rather than translated from it, since most chapter content (peripheral registers, memory maps, toolchain) is platform-specific. See that project’s README for background on why these are maintained as separate repositories rather than one.

Part I.

Foundations

1. Introduction

i Draft status

This chapter has not yet been written. It will cover what an embedded system and microcontroller are, why systems are embedded with computers, the constraints embedded systems operate under, and an overview of the STM32 development board used throughout this book.

2. C Programming

i Draft status

This chapter has not yet been written. It will cover low- vs. high-level languages, C programming basics, C and memory management, and using an STM32 IDE (e.g. STM32CubeIDE) to build and flash C programs.

Part II.

**Digital I/O and Program
Structure**

3. General Purpose Input/Output (GPIO)

i Draft status

This chapter has not yet been written. It will cover memory-mapped peripherals, the STM32 GPIO peripheral registers, controlling GPIO pins directly and via HAL/LL drivers, and bit masking.

4. Embedded Programming

i Draft status

This chapter has not yet been written. It will cover functionalizing code, hardware abstraction layers, software state machines, clock sources, and interrupts on STM32.

Part III.

Peripherals

5. Timers

i Draft status

This chapter has not yet been written. It will cover counters, timers, and the STM32 timer peripherals.

6. Analog-to-Digital Conversion (ADC)

i Draft status

This chapter has not yet been written. It will cover the STM32's analog-to-digital converter(s), configuring the ADC, sampling analog inputs, and reading conversion results.

7. Serial Communication

i Draft status

This chapter has not yet been written. It will cover serial communication peripherals on the STM32, such as USART, and how to use them to communicate with other devices.

